MIA PETKOVA

mia.petkova@gmail.com | www.linkedin.com/in/miapetkova | Denver, CO

PROFESSIONAL SUMMARY

Senior software/data engineer with 10+ years architecting cloud-native, production-scale data pipelines and infrastructure, with specialization in geospatial and earth-observation systems.

SKILLS

Airflow • AWS • Bash/Shell scripting • CI/CD • Docker • GDAL • GIS • Git • Image Processing (OpenCV) • Kubernetes • PDAL • PostgreSQL/PostGIS • Prefect • Python • Shapely/Fiona • SQL • Terraform • Xarray/Zarr

EXPERIENCE

Spire Global Inc. — Boulder, CO

SENIOR SOFTWARE ENGINEER, WEATHER — 2023 – Present

- Engineer production-scale data ingest and retrieval pipelines for weather and earth observation data (NetCDF, GRIB, GeoTIFF), powering mission-critical, customer-facing products
- Build customer-facing APIs (FastAPI) for global access to earth observation datasets
- Design and maintain AWS infrastructure provisioned with Terraform, leveraging ECS/Fargate and Step Functions to orchestrate scalable CPU-only and GPU-accelerated containerized workflows
- Migrated multiple workflow orchestration pipelines to a fully AWS-native architecture, reducing infrastructure costs by up to 80% and improving workflow performance by up to 90%

One Concern Inc — Boston, MA

SENIOR SOFTWARE ENGINEER — 2022 – 2022

- Built data pipelines (Argo) measuring infrastructure resilience across climate change scenarios
- Optimized pipeline throughput for high-volume ETL processes spanning GCP, Snowflake, and PostgreSQL
- Wrote unit tests for new pipeline components to ensure production reliability

AirWorks Solutions Inc — Boston, MA

COMPUTER VISION ENGINEER — 2018 – 2022

- Developed Python software and algorithms, leveraging GDAL and PDAL, for segmentation and feature detection in 2D aerial imagery and 3D LiDAR/point cloud data
- Built a containerized data-processing pipeline (AWS, Airflow, Docker) handling geospatial data across point cloud, raster, and vector formats
- Designed and led migration to a PostgreSQL/PostGIS-based storage and processing architecture, replacing legacy file-based storage and driving ~85% improvement in post-processing speed
- Maintained AWS cloud infrastructure and CI/CD pipelines for automated deployments across software suite
- As a founding engineer, hired, mentored, and provided technical guidance to junior engineers, helping establish team coding standards and practices

TellusLabs Inc — Somerville, MA

SOFTWARE ENGINEER — 2017 – 2017

- Built tools to collect, process, and visualize NASA satellite imagery for plant health analysis using Python, GDAL, PostgreSQL, Geoserver, and Airflow

Macro Connections Group, MIT Media Lab — Cambridge, MA

RESEARCH ASSISTANT — 2014 – 2017

- Developed StreetChange, a platform visualizing urban change using Google Street View data; Webby Award recipient for best use of Machine Learning

Civic Data Design Lab, MIT — Cambridge, MA

RESEARCH ASSISTANT — 2013 – 2017

McConnell Brain Imaging Centre, Montréal Neurological Institute — Montréal, QC, Canada

SOFTWARE DEVELOPER — 2009 – 2014

Georouting Technologies Inc. — Montréal, QC, Canada

GIS ANALYST — 2008 – 2009

EDUCATION

Massachusetts Institute of Technology — Cambridge, MA

MASTER OF SCIENCE, MEDIA ARTS & SCIENCES — 2013 – 2016

McGill University — Montréal, QC, Canada

CERTIFICATE, SOFTWARE DEVELOPMENT — 2009 – 2010

McGill University — Montréal, QC, Canada

BACHELOR OF ARTS, GEOGRAPHY [URBAN SYSTEMS] — 2004 – 2008